UPSSSC ALERT

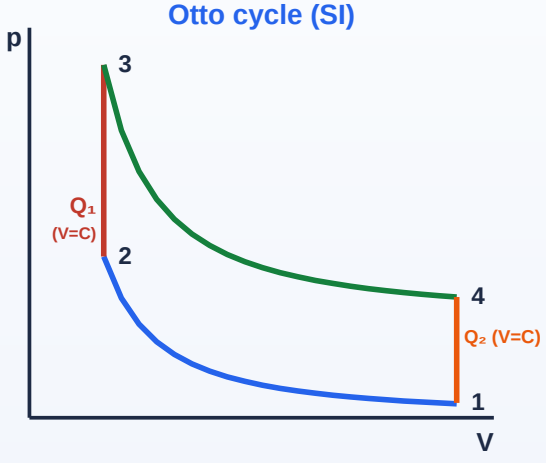
UPSSSC JE

MECHANICAL I.C. ENGINES

Complete English Edition

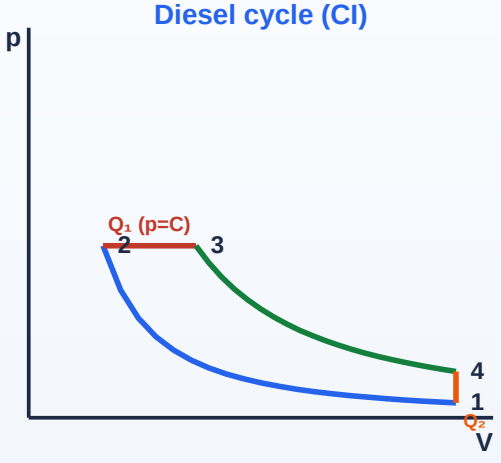
Unit 5 · 5 Marks · 2026

Otto cycle (SI)



$\eta = 1 - 1/r^{\gamma-1}$

Diesel cycle (CI)



$\eta = 1 - (1/\gamma r^{\gamma-1}) \cdot (\rho^{\gamma} - 1)/(\rho - 1)$

UPSSSC JE (Mechanical) 2026: Part-1 has **5 questions (5 marks)** from **I.C. Engines**. Syllabus: air-standard cycle, Diesel cycle and performance of IC engines. Basics and classification of IC engines are added for foundation.

Each chapter contains key points, diagrams, formula table, detailed explanation, solved numericals, 30 one-liners, MCQs with explanations and SSC JE PYQs. Content follows standard BTEUP diploma textbooks (V. Ganesan, R. K. Rajput, Mathur–Sharma).

For a better score, study this PDF and also attempt the QUIZ.

INDEX

S. No.	Topic	Page No.
1	**IC Engines — Basics & Classification**	02-11
2	**Air-Standard Cycles & Diesel Cycle**	12-21
3	**Performance of IC Engines**	22-31

UPSSSC ALERT

Chapter 1: IC Engines — Basics & Classification

IC Engines — Basics & Classification – Basic Concept

An **internal combustion (IC) engine** is a heat engine in which fuel burns inside the engine cylinder and the high-temperature, high-pressure gases push the piston to do work. Motorcycles, cars, trucks, tractors and diesel generators all use IC engines. In a steam engine combustion occurs outside (in a boiler), so it is an external combustion (EC) engine.

Understand in Detail — IC Engines — Basics & Classification

1.1 Heat engine and internal combustion engine — the basic idea

A **heat engine** is a machine that converts the chemical energy of fuel first into heat and then into mechanical work. If the fuel burns **outside** the engine (like the boiler of a steam engine), it is an **external combustion (EC)** engine. If the fuel burns **inside** the engine cylinder itself and the burning gases push the piston directly, it is an **internal combustion (IC)** engine — like a motorcycle, car, truck or tractor.

Advantages of IC engines: higher efficiency (no boiler losses), compact and light (less weight per kW), starts instantly. **Disadvantages:** cannot burn solid/cheap fuels (coal), vibration due to reciprocating motion, not suitable for very large power.

1.2 Engine parts and essential terminology

Main parts: cylinder (in which the piston moves), **cylinder head** (top cover holding valves, spark plug/injector), **piston** (receives the gas thrust), **piston rings** (compression rings — stop gas leakage; oil ring — stops oil entering the combustion chamber), **gudgeon pin** (joins piston to connecting rod), **connecting rod** (transmits the piston's straight motion to the crank), **crankshaft** (converts reciprocating motion into rotary motion), **flywheel** (energy store — smooths speed fluctuations), **camshaft** (opens/closes valves; in a four-stroke it runs at **half the crank speed**).

Terminology: TDC (top dead centre) — topmost piston position; **BDC (bottom dead centre)** — lowest. **Bore (D)** — inner diameter of the cylinder. **Stroke (L)** — distance moved by the piston from TDC to BDC = twice the crank radius. **Swept volume** $V_s = (\pi/4)D^2L$. **Clearance volume** V_c — volume left above the piston at TDC. **Compression ratio** $\mathbf{r = (V_c + V_s)/V_c}$ — petrol engines 6–10, diesel 14–22. **Mean piston speed** = 2LN/60 m/s.

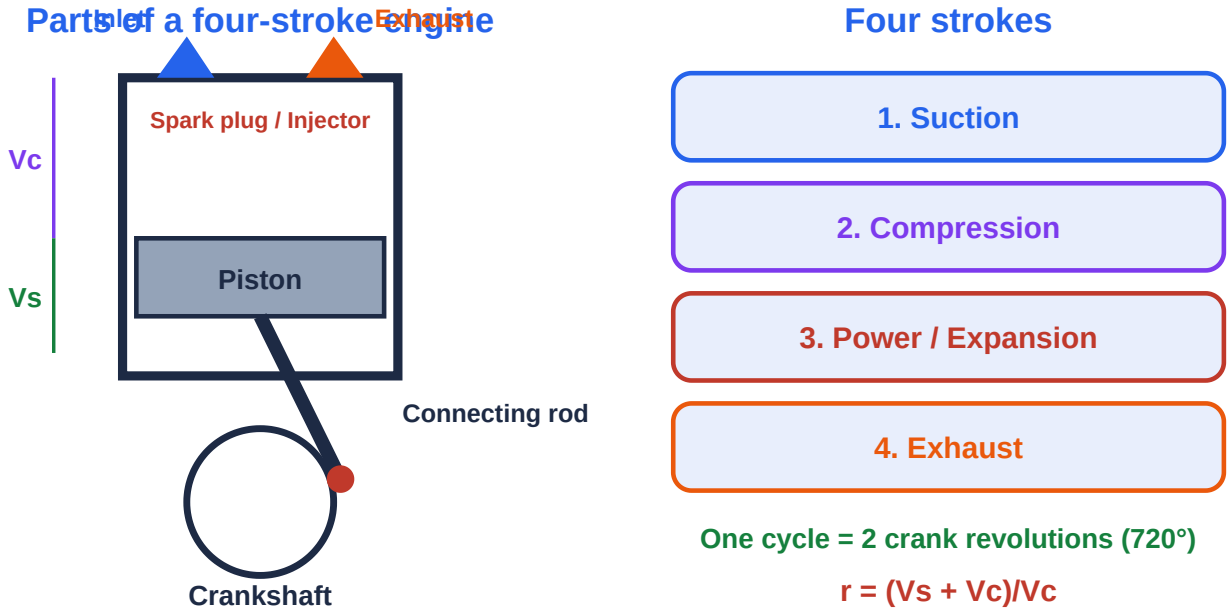


Fig 5.2: Main parts and strokes of a four-stroke engine

Example: An engine has bore 100 mm, stroke 120 mm and clearance volume 0.1 litre. Compression ratio?

Solution:

1. $V_s = (\pi/4)(0.1)^2(0.12) = 9.42 \times 10^{-4}$ m³ = 0.942 litre
2. r = (0.1 + 0.942)/0.1 = **10.42**

1.3 Four-stroke engine — watching one cycle slowly

In a four-stroke engine the piston makes four strokes to complete one working cycle — that is, **two revolutions of the crankshaft**. Power is obtained in only one of the four strokes, so a heavy flywheel is needed.

(1) Suction stroke: piston moves TDC → BDC; inlet valve open, exhaust closed. In a petrol (SI) engine an **air–petrol mixture** (from carburettor or injection) enters; in a diesel (CI) engine only **air**.

(2) Compression stroke: both valves closed, piston BDC → TDC; the charge is compressed and heats up. In SI, slightly before the end of the stroke the **spark plug** fires. In CI the air is compressed so much (r ≈ 18) that its temperature reaches about 600–700°C, and right then the **fuel injector** sprays fine diesel droplets that **ignite by themselves** in the hot air.

(3) Power (expansion) stroke: the very high pressure of burning gases pushes the piston towards BDC — **this is the only stroke that produces work**. **(4) Exhaust stroke:** exhaust valve open, piston moves up and pushes out the burnt gases.

Valve timing (real engine): valves do not open/close exactly at TDC/BDC. The inlet valve opens a little before TDC and closes after BDC (the inertia of incoming gas packs more charge); the exhaust valve opens before BDC (blow-down). For some degrees near TDC both valves are open — this is **valve overlap**; it helps sweep out residual gases (scavenging).

1.4 Two-stroke engine and a full comparison of SI vs CI

In a **two-stroke engine** the cycle is completed in **one crank revolution (two strokes)** — one power stroke every revolution. It usually has no valves; there are **ports** in the cylinder wall that the piston itself uncovers and covers. **Crankcase compression**: as the piston moves up, fresh charge is drawn into the crankcase below; as it moves down, this charge is slightly compressed and passes through the **transfer port** into the cylinder, pushing burnt gases out of the **exhaust port** — this is **scavenging**. A **deflector** on the piston crown prevents the fresh charge from escaping straight out of the exhaust.

Two-stroke vs four-stroke: two-stroke — theoretically twice the power for the same size (1.7–1.8 times in practice), lighter flywheel, simpler and cheaper; but part of the fresh charge escapes through the exhaust → **higher fuel consumption, more pollution**, and oil has to be mixed with petrol for lubrication. So it has disappeared from scooters/mopeds, but very large marine diesel engines still use two-stroke.

SI (petrol) vs CI (diesel): SI — **Otto cycle** (heat at constant volume), spark ignition, low compression ratio (6–10), **quantity governing** (the throttle reduces the quantity of mixture), light, high speed, low initial cost. CI — **Diesel cycle** (heat at constant pressure), auto-ignition by compression, high compression ratio (14–22), **quality governing** (air stays the same, only fuel is reduced), heavy and robust, higher efficiency, lower fuel cost.

Knocking — opposite in the two: in an SI engine, if the remaining mixture (end gas) auto-ignites **before** the spark-initiated flame reaches it, sharp pressure waves cause 'knock' — so SI fuel should have **delayed self-ignition → high octane number** (reference: iso-octane = 100, n-heptane = 0). In a CI engine, if the injected fuel ignites late (long **ignition delay**), a lot of fuel accumulates and burns at once, causing knock — so CI fuel should **ignite quickly → high cetane number** (reference: n-cetane = 100, α-methyl naphthalene = 0).

> **Note:** Factors that reduce knock in SI increase it in CI (and vice versa) — e.g. high compression ratio and high inlet temperature increase knock in SI but reduce it in CI.

1. Key Points

- **By ignition:** spark ignition (SI – petrol, Otto cycle, spark plug) and compression ignition (CI – diesel, Diesel cycle, fuel injector).
- **By strokes:** four-stroke (one power stroke per 2 revolutions) and two-stroke (one power stroke per revolution; ports instead of valves).
- **Compression ratio (r):** $r = (V_s + V_c)/V_c$. 6–10 for SI engines (knock limit), 14–22 for CI engines (for self-ignition).
- **Swept volume:** $V_s = (\pi/4)D^2L$; D = bore, L = stroke = 2 × crank radius.
- **Dead centres:** TDC (top dead centre) and BDC – extreme piston positions. Volume left at TDC = clearance volume (V_c).

- **Main parts:** cylinder, cylinder head, piston, piston rings (compression and oil rings), gudgeon pin, connecting rod, crankshaft, camshaft, valves, flywheel.
- **Valve timing:** the inlet valve opens before TDC and closes after BDC; the exhaust valve opens before BDC and closes after TDC; the period when both are open is valve overlap.
- **Camshaft speed:** in a four-stroke engine the camshaft runs at half the crankshaft speed.
- **Two-stroke engine:** scavenging (pushing out exhaust with fresh charge) occurs; lighter, simpler, higher power/weight ratio, but more fuel consumption and pollution.
- **Combustion abnormalities:** knocking (detonation) in SI engines — controlled by octane number; diesel knock in CI engines — controlled by cetane number.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Swept volume	$V_s = (\pi/4)\, D^2\, L$
Compression ratio	$r = (V_s + V_c)/V_c = 1 + V_s/V_c$
Stroke	**L = 2 × crank radius**
Mean piston speed	$V_p = 2LN/60$
Power strokes/min (4-stroke)	**N/2**
Power strokes/min (2-stroke)	**N**
SI engine CR	**6 – 10**
CI engine CR	**14 – 22**

3. Detailed Explanation

- **Working of four-stroke SI engine:** (1) suction – air-petrol mixture enters; (2) compression – mixture compressed, spark at end; (3) power – gases expand pushing the piston; (4) exhaust – burnt gases expelled.
- **Difference in CI engine:** only air is drawn in, high compression heats it to ~500–600°C, then the injector sprays diesel which self-ignites. Quantity governing.
- **Carburettor/MPFI in SI engines:** prepares the air-fuel mixture; throttle controls the quantity of mixture — quantity governing in SI, quality governing in CI.
- **Cooling and lubrication:** air cooling (fins – motorcycles) or water cooling (radiator – cars); lubrication – splash, forced (pressure) and petroil (oil mixed with petrol in two-strokes).
- **Fuel quality:** octane number — reference mixture of iso-octane (100) and n-heptane (0); cetane number — cetane (100) and α-methyl naphthalene (0).

4. Comparison Table — SI vs CI Engine

Basis	SI (Petrol)	CI (Diesel)
Cycle	Otto (heat at V = C)	Diesel (heat at p = C)
Ignition	Spark plug	Self-ignition by compression
Compression ratio	6–10	14–22
Drawn in	Air + fuel	Air only
Governing	Quantity	Quality
Weight / speed	Lighter, high speed	Heavier, low speed
Fuel rating	Octane number	Cetane number

4. Comparison Table — Four-stroke vs Two-stroke

Basis	Four-stroke	Two-stroke
Power stroke	1 in 2 revolutions	1 every revolution
Valves / ports	Valves	Ports
Flywheel	Heavier	Lighter
Efficiency	Higher	Lower

5. Solved Numerical Problems

Example 1. Bore 100 mm, stroke 120 mm and clearance volume 0.125 litre. Find the compression ratio.

Solution:

1. $V_s = (\pi/4) \times 10^2 \times 12 = 942.5\ cm^3$
2. $V_c = 125\ cm^3$
3. r = (942.5 + 125)/125 = **8.54**

Example 2. A four-stroke engine runs at 3000 rpm. Find power strokes per minute and camshaft speed.

Solution:

1. Power strokes = N/2 = **1500 per minute**
2. Camshaft = 3000/2 = **1500 rpm**

Example 3. Stroke 100 mm, speed 1800 rpm. Find the mean piston speed.

Solution:

1. $V_p = 2 \times 0.1 \times 1800/60$ = **6 m/s**

Example 4. Compression ratio 16 and swept volume 1500 cm³. Find the clearance volume.

Solution:

1. $V_c = V_s/(r - 1) = 1500/15 =$ **100 cm³**

6. Frequently Asked Facts

- SI – Otto cycle, spark plug, r = 6–10; CI – Diesel cycle, injector, r = 14–22.
- Four-stroke: one power stroke in 2 revolutions; camshaft at half speed.
- Octane number – petrol; cetane number – diesel.
- Quantity governing in SI, quality governing in CI.

7. Note (Confusing Facts)

- A diesel engine has no spark plug; fuel ignites in hot air.
- Compression ratio is a ratio of volumes, not pressures.
- A two-stroke engine does not give double power (scavenging losses), only about 1.5–1.7 times.

Remember: One line: $r = 1 + V_s/V_c$. Assertion–Reason: (A) Diesel engines have a high compression ratio. (R) Air must be very hot for the fuel to self-ignite. — Both true; R explains A.

30 Most Important One-Liner Facts — IC Engines — Basics & Classification

1. Where is combustion in IC engine?

— Inside the cylinder.

2. Steam engine is?

— External combustion.

3. Petrol engine cycle?

— Otto.

4. Diesel engine cycle?

— Diesel.

5. Ignition in SI engine?

— Spark plug.

6. Ignition in CI engine?

— Self-ignition by compression.

7. CR of SI engine?

— 6–10.

8. CR of CI engine?

— 14–22.

9. Compression ratio formula?

— $(V_s + V_c)/V_c$.

10. Swept volume?

— $(\pi/4)D^2L$.

11. Stroke and crank radius?

— $L = 2r$.

12. TDC stands for?

— Top Dead Centre.

13. Volume at TDC?

— Clearance volume.

14. One cycle in four-stroke = ?

— 2 revolutions (720°).

15. One cycle in two-stroke = ?

— 1 revolution (360°).

16. Camshaft speed in four-stroke?

— Half crank speed.

17. Instead of valves in two-stroke?

— Ports.

18. Removing exhaust with fresh charge?

— Scavenging.

19. Both valves open — period?

— Valve overlap.

20. Petrol quality rating?

— Octane number.

21. Diesel quality rating?

— Cetane number.

22. Abnormal combustion in SI?

— Knocking (detonation).

23. Governing of SI engine?

— Quantity.

24. Governing of CI engine?

— Quality.

25. Pin joining piston and rod?

— Gudgeon pin.

26. Lubrication of 2-stroke petrol?

— Petroil.

27. Motorcycle engine cooling?

— Air (fins).

28. Mean piston speed?

— 2LN/60.

29. Drawn in CI engine?

— Only air.

30. Octane reference fuel (100)?

— Iso-octane.

Practice MCQs — IC Engines — Basics & Classification

Q1. Ignition in a diesel engine is by:

(a) Spark plug
(b) Heat of compression
(c) Glow plug always
(d) Magneto

✔ **Answer: (b) Heat of compression**

Explanation: High compression heats the air.

Q2. One cycle of a four-stroke engine takes:

(a) 1 revolution
(b) 2 revolutions
(c) 4 revolutions
(d) ½ revolution

✔ **Answer: (b) 2 revolutions**

Explanation: 720°.

Q3. Compression ratio of a petrol engine:

(a) 2–4
(b) 6–10
(c) 14–22
(d) 25–30

✔ **Answer: (b) 6–10**

Explanation: Knock limit.

Q4. Compression ratio is:

(a) V_s/V_c
(b) $(V_s + V_c)/V_c$
(c) V_c/V_s
(d) $(V_s + V_c)/V_s$

✔ **Answer: (b) $(V_s + V_c)/V_c$**

Explanation: Total ÷ clearance volume.

Q5. Quality of diesel fuel is rated by:

(a) Octane number
(b) Cetane number
(c) Flash point
(d) Viscosity

✔ **Answer: (b) Cetane number**

Explanation: Self-ignition quality.

Q6. Camshaft speed in a four-stroke engine:

(a) Equal to crank
(b) Half of crank
(c) Twice crank
(d) Quarter of crank

✔ **Answer: (b) Half of crank**

Explanation: Each valve opens once per 2 revolutions.

Q7. Power strokes per revolution in a two-stroke engine:

(a) ½
(b) 1
(c) 2
(d) 4

✔ **Answer: (b) 1**

Explanation: Every revolution.

Q8. Drawn in during suction in an SI engine:

(a) Air only
(b) Air–fuel mixture
(c) Fuel only
(d) Exhaust gas

✔ **Answer: (b) Air–fuel mixture**

Explanation: Mixture from carburettor/injection.

Q9. Governing of a CI engine is:

(a) Quantity
(b) Quality
(c) Hit and miss
(d) Combined

✔ **Answer: (b) Quality**

Explanation: Only fuel quantity changes, not air.

Q10. $V_s = 1500\ cm^3$, $r = 16$ — clearance volume:

(a) 93.75 cm^3
(b) 100 cm^3
(c) 1600 cm^3
(d) 150 cm^3

✔ **Answer: (b) 100 cm^3**

Explanation: $1500/(16 - 1)$.

Q11. Pushing out burnt gases with fresh charge:

(a) Throttling
(b) Scavenging
(c) Supercharging
(d) Knocking

✔ **Answer: (b) Scavenging**

Explanation: A two-stroke process.

Q12. Knocking in petrol engines is reduced by:

(a) Higher octane fuel
(b) Higher compression ratio
(c) Lower cetane
(d) Higher intake temperature

✔ **Answer: (a) Higher octane fuel**

Explanation: Octane number is anti-knock quality.

Q13. The gudgeon pin connects:

(a) Piston and connecting rod
(b) Crank and rod
(c) Valve and cam
(d) Cylinder and head

✔ **Answer: (a) Piston and connecting rod**

Explanation: Small end.

Q14. Stroke 100 mm, 1800 rpm — mean piston speed:

(a) 3 m/s
(b) 6 m/s
(c) 9 m/s
(d) 12 m/s

✔ **Answer: (b) 6 m/s**

Explanation: 2 × 0.1 × 30.

Q15. Octane reference fuel rated 0:

(a) Iso-octane
(b) n-heptane
(c) Cetane
(d) Benzene

✔ **Answer: (b) n-heptane**

Explanation: n-heptane = 0.

SSC JE / UPSSSC JE Previous Year Questions — IC Engines — Basics & Classification

Q1. Which part is absent in a diesel engine?

SSC JE (Previous Year)

(a) Fuel injector
(b) Spark plug
(c) Crankshaft
(d) Flywheel

✔ **Answer: (b) Spark plug**

Explanation: Self-ignition.

Q2. Purpose of scavenging in a two-stroke engine:

SSC JE (Previous Year)

(a) Save fuel
(b) Remove burnt gases
(c) Cooling
(d) Lubrication

✔ **Answer: (b) Remove burnt gases**

Explanation: Expelling with fresh charge.

Q3. Compression ratio of a CI engine is usually:

SSC JE (Previous Year)

(a) 5–8
(b) 8–10
(c) 14–22
(d) 30–40

✔ **Answer: (c) 14–22**

Explanation: For self-ignition.

Q4. Ideal cycle of a petrol engine:

SSC JE (Previous Year)

(a) Diesel
(b) Otto
(c) Rankine
(d) Brayton

✔ **Answer: (b) Otto**

Explanation: Heat at constant volume.

Q5. In a four-stroke engine each valve opens once per:

SSC JE (Previous Year)

(a) 1 revolution
(b) 2 revolutions
(c) 4 revolutions
(d) ½ revolution

✔ **Answer: (b) 2 revolutions**

Explanation: 720°.

Q6. Octane number is related to:

SSC JE (Previous Year)

(a) Diesel
(b) Petrol
(c) Kerosene
(d) Lubricating oil

✔ **Answer: (b) Petrol**

Explanation: SI fuel.

Chapter 2: Air-Standard Cycles & Diesel Cycle

Air-Standard Cycles & Diesel Cycle – Basic Concept

A real engine involves combustion, friction and heat loss, making analysis difficult. In an **air-standard cycle** the working fluid is taken as air (ideal gas), combustion is replaced by external heat addition and all processes are reversible, to obtain a theoretical efficiency. The Otto (SI), Diesel (CI) and Dual cycles are the main ones.

Understand in Detail — Air-Standard Cycles & Diesel Cycle

2.1 Air-standard cycles — understanding the engine by simplifying it

In a real engine many things happen at once — fuel burns, gas composition changes, heat escapes through the walls. To simplify analysis we take some **ideal assumptions**: (1) the working substance is only **air**, behaving as an ideal gas; (2) specific heats (c_p, c_v) are constant; (3) instead of combustion **heat is supplied** from an external source, and instead of exhaust heat is rejected; (4) compression and expansion are **reversible adiabatic**. Such a cycle is an **air-standard cycle** and its efficiency the **air-standard efficiency** — it gives the upper limit for a real engine.

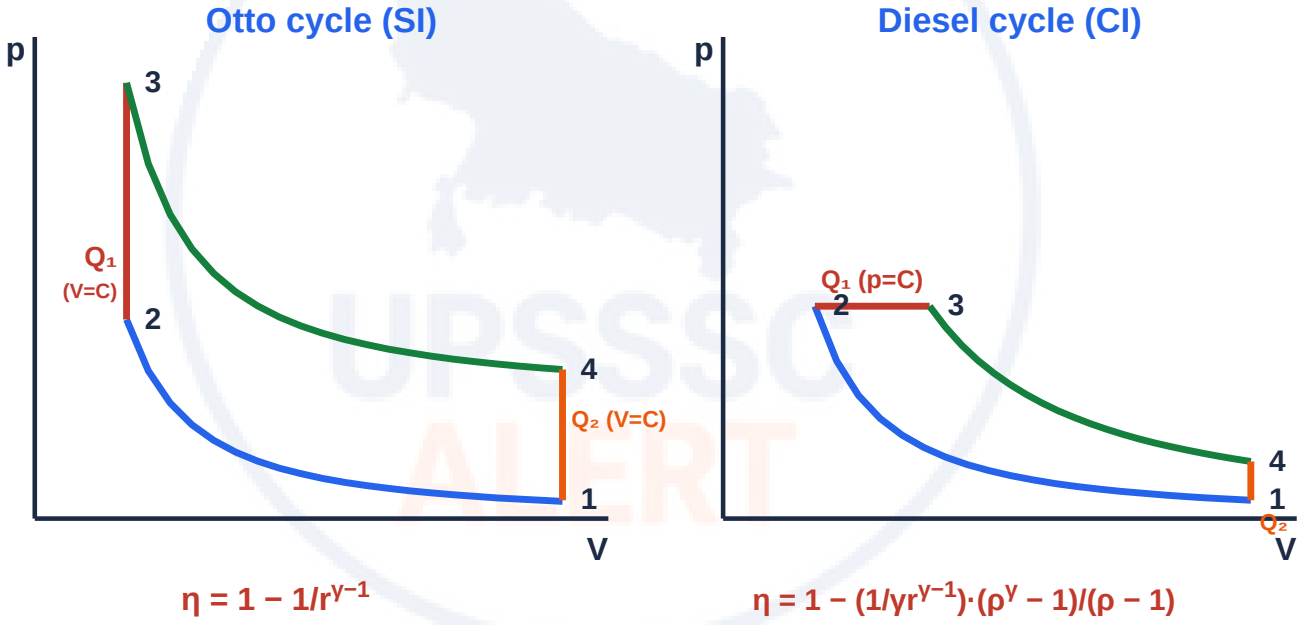


Fig 5.1: p–V diagrams of Otto and Diesel cycles

2.2 Otto cycle — derivation and meaning

The Otto cycle (1876, Nikolaus Otto) is the ideal cycle of the petrol engine. Four processes: **1→2** adiabatic compression; **2→3** heat addition at **constant volume** (spark combustion is so fast that the piston hardly moves); **3→4** adiabatic expansion; **4→1** heat rejection at constant volume.

Derivation: Otto efficiency

1. $Q_{in} = m\,c_v\,(T_3 - T_2)$; $Q_{out} = m\,c_v\,(T_4 - T_1)$
2. $\eta = 1 - Q_{out}/Q_{in} = 1 - (T_4 - T_1)/(T_3 - T_2)$
3. Adiabatic: $T_2/T_1 = r^{\gamma-1}$ and $T_3/T_4 = r^{\gamma-1}$ (both have volume ratio r)
4. So $T_2 = T_1 r^{\gamma-1}$, $T_3 = T_4 r^{\gamma-1}$ → $T_3 - T_2 = (T_4 - T_1)\,r^{\gamma-1}$
5. $\boldsymbol{\eta_{Otto} = 1 - 1/r^{\gamma-1}}$ — depends only on compression ratio r and γ, not on the amount of heat supplied

Meaning: increasing the compression ratio increases efficiency. Then why not r = 20 in a petrol engine? Because at high r the temperature at the end of compression rises so much that the mixture ignites by itself before the spark — **knocking**. That is why r of SI engines is limited to 6–10, and high-octane petrol allows a slightly higher r. **Mean effective pressure (MEP)** = net work ÷ swept volume — the constant pressure which, acting over the whole expansion stroke, would do the same work.

Example: Otto cycle with r = 8, γ = 1.4. Air-standard efficiency?

Solution:

1. $r^{\gamma-1} = 8^{0.4} = 2.297$
2. $\eta = 1 - 1/2.297 = 1 - 0.435 =$ **56.5%**
3. If $T_1 = 300$ K, end-of-compression temperature $T_2 = 300 \times 2.297 = 689$ K (416°C)

2.3 Diesel cycle and dual cycle — the role of the cut-off ratio

In the Diesel cycle (1897, Rudolf Diesel) heat is supplied at **constant pressure** — because diesel comes gradually from the injector and keeps burning while the piston moves down, so the pressure remains nearly constant. Processes: 1→2 adiabatic compression; **2→3 heat addition at constant pressure**; 3→4 adiabatic expansion; 4→1 heat rejection at constant volume.

Cut-off ratio (ρ) = V_3/V_2 — volume when fuel burning stops ÷ volume at end of compression. It tells how long the fuel kept burning, i.e. how much load the engine carries.

Derivation: Diesel efficiency

1. $Q_{in} = m\,c_p(T_3 - T_2)$; $Q_{out} = m\,c_v(T_4 - T_1)$ → $\eta = 1 - (T_4 - T_1)/[\gamma(T_3 - T_2)]$
2. $T_2 = T_1 r^{\gamma-1}$; $T_3 = \rho T_2$ (constant pressure) ; $T_4 = T_1 \rho^{\gamma}$ (from expansion and compression)
3. $\boldsymbol{\eta_{Diesel} = 1 - (1/r^{\gamma-1}) \times [(\rho^{\gamma} - 1)/(\gamma(\rho - 1))]}$
4. The bracketed term is always greater than 1 ($\rho > 1$) → **for the same r, Diesel efficiency < Otto efficiency**; as ρ increases (more load) Diesel efficiency falls

Then why is the diesel engine more efficient? Because in reality a diesel engine has a very high r (16–22), while a petrol engine has 8–10 — the higher r gives far more benefit than the cut-off loss. **Comparison rules:** (1) for the **same compression ratio and heat input**: $\eta_{Otto} > \eta_{Dual} > \eta_{Diesel}$. (2) for the **same maximum pressure and temperature** (the real limit): $\eta_{Diesel} > \eta_{Dual} > \eta_{Otto}$.

Dual / mixed cycle (Sabathe / limited pressure): in modern high-speed diesel engines some fuel burns at constant volume and the rest at constant pressure. It has both a pressure ratio $\alpha = p_3/p_2$ and a cut-off ρ. $\alpha = 1 \rightarrow$ Diesel; $\rho = 1 \rightarrow$ Otto.

> **Example:** Diesel cycle with r = 16, $\rho = 2$, $\gamma = 1.4$. Efficiency?
>
> **Solution:**
>
> 1. $16^{0.4} = 3.031$; $\rho^{\gamma} = 2^{1.4} = 2.639$
> 2. Bracket = $(2.639 - 1)/(1.4 \times 1) = 1.171$
> 3. $\eta = 1 - 1.171/3.031 = 1 - 0.386 =$ **61.4%** (Otto at the same r = 16: $1 - 1/3.031 = 67\%$)

1. Key Points

- **Air-standard assumptions:** working fluid is air (ideal gas with constant c_p, c_v); closed cycle; compression and expansion reversible adiabatic; heat added and rejected externally.
- **Otto cycle:** 1–2 isentropic compression, 2–3 heat addition at constant volume, 3–4 isentropic expansion, 4–1 heat rejection at constant volume. $\eta = 1 - 1/r^{\gamma-1}$ — depends only on r.
- **Diesel cycle:** 1–2 isentropic compression, 2–3 heat addition at constant pressure, 3–4 isentropic expansion, 4–1 heat rejection at constant volume. $\eta = 1 - (1/\gamma r^{\gamma-1}) \cdot (\rho^{\gamma} - 1)/(\rho - 1)$.
- **Cut-off ratio (ρ):** $\rho = V_3/V_2$ — the period of fuel injection. As ρ increases (higher load) the Diesel cycle efficiency decreases.
- **Dual (limited pressure) cycle:** part of the heat is added at constant volume and part at constant pressure; closest ideal for modern high-speed diesel engines.
- **Comparison (same compression ratio and heat input):** $\eta_{Otto} > \eta_{Dual} > \eta_{Diesel}$.
- **Comparison (same maximum pressure and temperature):** $\eta_{Diesel} > \eta_{Dual} > \eta_{Otto}$ — which is why diesel engines are more efficient in practice.
- **Mean effective pressure (MEP):** the constant pressure that, acting over the whole stroke, does the same work as the cycle: p_m = work/swept volume.
- **Air-standard vs actual:** the indicated thermal efficiency of a real engine is about 50–70% of the air-standard efficiency (relative efficiency).

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Otto efficiency	$\eta = 1 - 1/r^{\gamma-1}$
Diesel efficiency	$\eta = 1 - (1/\gamma r^{\gamma-1})\cdot(\rho^{\gamma} - 1)/(\rho - 1)$
Cut-off ratio	$\rho = V_3/V_2$
Isentropic relation	$T_2/T_1 = r^{\gamma-1}$, $p_2/p_1 = r^{\gamma}$
Heat added (Otto)	$Q = m\, c_v(T_3 - T_2)$
Heat added (Diesel)	$Q = m\, c_p(T_3 - T_2)$
Mean effective pressure	$p_m = W / V_s$
Relative efficiency	$\eta_{rel} = \eta_{actual}/\eta_{air\ std}$
Comparison (same r)	**Otto > Dual > Diesel**

3. Detailed Explanation

- **Derivation of Otto efficiency:** $Q_{in} = c_v(T_3 - T_2)$, $Q_{out} = c_v(T_4 - T_1)$; $\eta = 1 - (T_4 - T_1)/(T_3 - T_2) = 1 - T_1/T_2 = 1 - 1/r^{\gamma-1}$.
- **Effect of compression ratio:** efficiency rises with r, but in SI engines r is limited by knocking; CI engines have high r and thus higher efficiency.
- **Effect of γ:** efficiency increases with γ; a monatomic gas (γ = 1.67) would give more than air (γ = 1.4).
- **Diesel cycle with ρ = 1:** the Diesel efficiency equals the Otto efficiency — heat addition becomes instantaneous.
- **Actual vs ideal cycle:** variable specific heats, dissociation, heat loss, time loss (combustion not instantaneous), pumping loss, exhaust blow-down.

4. Comparison Table — Otto, Diesel and Dual Cycles

Basis	Otto	Diesel	Dual
Heat addition	V = C	p = C	V = C and p = C
Heat rejection	V = C	V = C	V = C
Efficiency depends on	r, γ	r, ρ, γ	r, ρ, α, γ
Order for same r	Highest	Lowest	Middle
Engine	Petrol	Slow-speed diesel	High-speed diesel

5. Solved Numerical Problems

Example 1. An Otto cycle has compression ratio 8 ($\gamma = 1.4$). Find the air-standard efficiency.

Solution:

1. $r^{\gamma-1} = 8^{0.4} = 2.297$
2. $\eta = 1 - 1/2.297 =$ **56.5%**

Example 2. A Diesel cycle has $r = 16$, $\rho = 2$, $\gamma = 1.4$. Find the efficiency.

Solution:

1. $r^{\gamma-1} = 16^{0.4} = 3.031$
2. $(\rho^{\gamma} - 1)/(\rho - 1) = (2^{1.4} - 1)/1 = 1.639$
3. $\eta = 1 - (1/(1.4 \times 3.031)) \times 1.639 = 1 - 0.386 =$ **61.4%**

Example 3. Otto cycle: $T_1 = 300$ K, $r = 8$ ($\gamma = 1.4$). Find the temperature after compression.

Solution:

1. $T_2 = T_1 \times r^{\gamma-1} = 300 \times 2.297 =$ **689 K**

Example 4. An engine has air-standard efficiency 55% and indicated thermal efficiency 33%. Find the relative efficiency.

Solution:

1. $\eta_{rel} = 33/55 =$ **60%**

Example 5. A cycle produces 600 J per cycle with a swept volume of 1 litre. Find the MEP.

Solution:

1. $p_m = 600/0.001 = 6 \times 10^5$ Pa = **6 bar**

6. Frequently Asked Facts

- Otto efficiency depends only on r (and γ): $\eta = 1 - 1/r^{\gamma-1}$.
- Same r: Otto > Dual > Diesel; same max pressure: Diesel > Dual > Otto.
- Diesel cycle adds heat at constant pressure; efficiency falls as cut-off ratio rises.
- Both cycles reject heat at constant volume.

7. Note (Confusing Facts)

- At the same r Otto is more efficient, yet diesel engines are more efficient in practice — because their r is much higher.
- With $\rho = 1$, Diesel = Otto.
- In the Diesel cycle heat is rejected at constant volume, not constant pressure.

Remember: One line: Otto – V = C, Diesel – p = C, Dual – both. Assertion–Reason: (A) Diesel cycle efficiency falls with load. (R) Cut-off ratio increases with load. — Both true; R explains A.

30 Most Important One-Liner Facts — Air-Standard Cycles & Diesel Cycle

1. Working fluid in air-standard cycle?

— Air (ideal gas).

2. Heat addition in Otto?

— Constant volume.

3. Heat addition in Diesel?

— Constant pressure.

4. Heat rejection in both?

— Constant volume.

5. Otto efficiency?

— $1 - 1/r^{\gamma-1}$.

6. Otto efficiency depends on?

— Only r (and γ).

7. Cut-off ratio?

— V_3/V_2.

8. Diesel efficiency as ρ rises?

— Decreases.

9. Diesel efficiency at ρ = 1?

— Equal to Otto.

10. Most efficient for same r?

— Otto.

11. Most efficient for same max pressure?

— Diesel.

12. Cycle of high-speed diesel?

— Dual.

13. Other name of dual cycle?

— Limited pressure cycle.

14. T_2/T_1 in isentropic?

— $r^{\gamma-1}$.

15. p_2/p_1 in isentropic?

— r^{γ}.

16. Otto efficiency at r = 8?

— ≈ 56.5%.

17. Mean effective pressure?

— Work ÷ swept volume.

18. Relative efficiency?

— Actual ÷ air-standard efficiency.

19. Efficiency as γ rises?

— Increases.

20. Otto efficiency as r rises?

— Increases.

21. Why r limited in SI?

— Knocking.

22. Processes in Otto?

— 2 isentropic + 2 constant volume.

23. Processes in Diesel?

— 2 isentropic + 1 p = C + 1 V = C.

24. Heat added in Otto?

— $mc_v(T_3 - T_2)$.

25. Heat added in Diesel?

— $mc_p(T_3 - T_2)$.

26. Why actual efficiency lower?

— Heat loss, friction, variable specific heats.

27. Gas turbine cycle?

— Brayton (Joule).

28. Steam plant cycle?

— Rankine.

29. Otto vs Carnot?

— Less efficient.

30. Area on p–V shows?

— Work per cycle.

Practice MCQs — Air-Standard Cycles & Diesel Cycle

Q1. In the Otto cycle heat is added at:

(a) Constant pressure
(b) Constant volume
(c) Constant temperature
(d) Constant entropy

✔ **Answer: (b) Constant volume**

Explanation: V = C.

Q2. Efficiency of the Otto cycle:

(a) $1 - 1/r^{\gamma}$
(b) $1 - 1/r^{\gamma-1}$
(c) $1 - r^{\gamma-1}$
(d) $1/r^{\gamma-1}$

✔ **Answer: (b) $1 - 1/r^{\gamma-1}$**

Explanation: Standard.

Q3. Heat addition in the Diesel cycle:

(a) Constant volume
(b) Constant pressure
(c) Constant temperature
(d) Adiabatic

✔ **Answer: (b) Constant pressure**

Explanation: p = C.

Q4. Correct order for the same compression ratio:

(a) Otto > Dual > Diesel
(b) Diesel > Dual > Otto
(c) Dual > Otto > Diesel
(d) Otto = Diesel

✔ **Answer: (a) Otto > Dual > Diesel**

Explanation: Standard comparison.

Q5. As cut-off ratio increases, Diesel efficiency:

(a) Increases
(b) Decreases
(c) Constant
(d) First rises then falls

✔ **Answer: (b) Decreases**

Explanation: $(\rho^{\gamma} - 1)/\gamma(\rho - 1)$ increases.

Q6. Otto efficiency at r = 8, γ = 1.4 is about:

(a) 43%
(b) 56.5%
(c) 65%
(d) 70%

✔ **Answer: (b) 56.5%**

Explanation: 1 − 1/2.297.

Q7. Otto efficiency depends on:

(a) Only compression ratio
(b) Cut-off ratio
(c) Maximum temperature
(d) Fuel

✔ **Answer: (a) Only compression ratio**

Explanation: (γ taken constant).

Q8. High-speed diesel engines approximate:

(a) Otto
(b) Diesel
(c) Dual
(d) Carnot

✔ **Answer: (c) Dual**

Explanation: Both types of heat addition.

Q9. With ρ = 1, Diesel cycle efficiency:

(a) Zero
(b) Equals Otto
(c) 100%
(d) Equals Carnot

✔ **Answer: (b) Equals Otto**

Explanation: Instantaneous heat addition.

Q10. Mean effective pressure =

(a) Work × swept volume
(b) Work ÷ swept volume
(c) Max pressure ÷ 2
(d) Average of pressures

✔ **Answer: (b) Work ÷ swept volume**

Explanation: $p_m = W/V_s$.

Q11. In isentropic compression T_2/T_1 =

(a) r
(b) r^{γ}
(c) $r^{\gamma-1}$
(d) $r^{1/\gamma}$

✔ **Answer: (c) $r^{\gamma-1}$**

Explanation: $Tv^{\gamma-1} = C$.

Q12. Most efficient for the same maximum pressure:

(a) Otto
(b) Diesel
(c) Dual
(d) All equal

✔ **Answer: (b) Diesel**

Explanation: Diesel has higher r.

Q13. Heat rejection in the Diesel cycle:

(a) Constant pressure
(b) Constant volume
(c) Constant temperature
(d) Constant enthalpy

✔ **Answer: (b) Constant volume**

Explanation: 4–1 at V = C.

Q14. 600 J/cycle, V_s = 1 litre — MEP:

(a) 0.6 bar
(b) 6 bar
(c) 60 bar
(d) 600 bar

✔ **Answer: (b) 6 bar**

Explanation: 600/0.001 Pa.

Q15. Compression ratio of SI engines is limited by:

(a) Knocking
(b) Friction
(c) Cooling
(d) Weight

✔ **Answer: (a) Knocking**

Explanation: Knocking at high r.

SSC JE / UPSSSC JE Previous Year Questions — Air-Standard Cycles & Diesel Cycle

Q1. In the Diesel cycle heat is supplied at:

SSC JE (Previous Year)

(a) Constant volume
(b) Constant pressure
(c) Constant temperature
(d) Constant entropy

✔ **Answer: (b) Constant pressure**

Explanation: p = C.

Q2. For the same compression ratio, Otto efficiency compared with Diesel is:

SSC JE (Previous Year)

(a) Higher
(b) Lower
(c) Equal
(d) Uncertain

✔ **Answer: (a) Higher**

Explanation: Otto > Diesel.

Q3. The Otto cycle consists of:

SSC JE (Previous Year)

(a) Two isentropic and two constant volume processes
(b) Two isothermal and two isentropic
(c) Two constant pressure and two isentropic
(d) Two isothermal and two constant volume

✔ **Answer: (a) Two isentropic and two constant volume processes**

Explanation: Otto.

Q4. Diesel cycle efficiency increases with:

SSC JE (Previous Year)

(a) Increase in cut-off ratio
(b) Increase in compression ratio
(c) Decrease in γ
(d) Increase in load

✔ **Answer: (b) Increase in compression ratio**

Explanation: With increase in r.

Q5. Heat addition in the dual cycle is at:

SSC JE (Previous Year)

(a) Constant volume only

(b) Constant pressure only

(c) Both constant volume and constant pressure

(d) Constant temperature

✔ **Answer: (c) Both constant volume and constant pressure**

Explanation: Both.

UPSSSC ALERT

Chapter 3: Performance of IC Engines

Chapter 3 · Performance of IC Engines

Performance of IC Engines – Basic Concept

The **performance** of an engine shows how efficiently it converts fuel energy into useful work. Parameters such as indicated power, brake power, friction power, mechanical efficiency, thermal efficiency, volumetric efficiency and specific fuel consumption are used.

Understand in Detail — Performance of IC Engines

3.1 Engine power — indicated, brake and friction

Engine power can be measured at two places: how much work the gases do on the piston **inside** the cylinder — **indicated power (IP)**; and how much useful power is actually obtained **outside** at the crankshaft — **brake power (BP)**. The difference is consumed in friction inside the engine (piston rings, bearings), pumping and accessories (fan, pump) — **friction power (FP) = IP − BP**.

Derivation: Formula for indicated power

1. Area of indicator diagram ÷ its length × spring constant = **mean effective pressure p_m**
2. Work per power stroke = force × distance = (p_m × A) × L
3. Power strokes per minute n = N/2 (four-stroke) or N (two-stroke) ; k cylinders
4. **IP = p_m L A n k / 60** (W, p_m in Pa) — memory trick: **'PLANK'**

Brake power is measured with a **dynamometer** — a brake is applied on the engine shaft and torque T is measured: **BP = 2πNT/60**. In a rope-brake dynamometer T = (W − S)(D + d)/2, where W = dead load, S = spring balance reading. Methods of measuring **friction power**: **Morse test** (cutting off one cylinder at a time in a multi-cylinder engine — only for multi-cylinder), **Willan's line** (extrapolating the graph of fuel consumption vs BP backward — only for diesel), motoring test, retardation test.

3.2 Efficiencies, specific fuel consumption and heat balance

Mechanical efficiency η_m = BP/IP — how much of the power developed inside the engine is delivered outside (≈ 80–90%). **Indicated thermal efficiency = IP/($\dot{m}_f$ × CV)**. **Brake thermal efficiency = BP/($\dot{m}_f$ × CV)** — fraction of total fuel energy obtained as useful power (petrol ≈ 25–30%, diesel ≈ 30–40%). **Relative efficiency** = actual thermal efficiency ÷ air-standard efficiency.

Volumetric efficiency = volume of air actually inducted (at atmospheric conditions) ÷ swept volume — tells how well the engine 'breathes' (≈ 70–90%). With a supercharger/turbocharger pushing in pre-compressed air it can even exceed 100%.

Brake specific fuel consumption (BSFC) = fuel per hour ÷ BP, kg/kWh — **the lower, the more economical the engine**. Brake thermal efficiency = 3600/(BSFC × CV). A diesel engine's BSFC is about 0.2–0.25 kg/kWh.

Heat balance sheet: where did the 100% heat supplied by fuel go? Roughly — **brake power ≈ 30%, cooling water ≈ 30%, exhaust gases ≈ 30%, radiation and others ≈ 10%**. This shows nearly two-thirds of the energy is wasted as heat — which is why a turbocharger (using exhaust energy) is so beneficial.

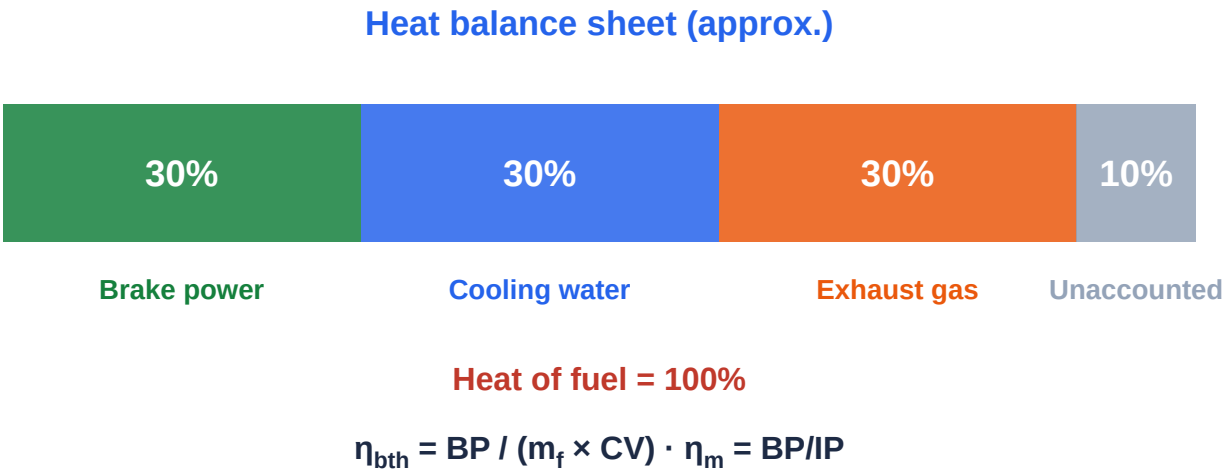


Fig 5.3: Heat balance of an IC engine

Example: Four-stroke single-cylinder engine: bore 0.2 m, stroke 0.3 m, p_m = 7 bar, N = 300 rpm, BP = 13 kW, fuel 6 kg/h, CV = 42000 kJ/kg. Find IP, η_m, brake thermal efficiency and BSFC.

Solution:

1. A = $(\pi/4)(0.2)^2$ = 0.03142 m² ; n = 300/2 = 150
2. IP = $7 \times 10^5 \times 0.3 \times 0.03142 \times 150/60$ = 16493 W ≈ **16.5 kW**
3. η_m = BP/IP = 13/16.5 = **78.8%**
4. Fuel energy = (6/3600) × 42000 = 70 kW → η_{bth} = 13/70 = **18.6%**
5. BSFC = 6/13 = **0.46 kg/kWh** ; friction power = 16.5 − 13 = 3.5 kW — check: BP must always be less than IP

3.3 Fuel system, ignition, cooling and lubrication — brief but clear

SI fuel system: older engines use a **carburettor** — at the narrow throat of a venturi the air velocity rises and pressure falls, drawing petrol from the float chamber and atomising it into the air. Today **MPFI** (multi-point fuel injection) — electronically controlled precise quantity at each cylinder inlet. **Chemically correct (stoichiometric) air-fuel ratio ≈ 14.7 : 1**; a cold engine needs a rich mixture to start — that is what the **choke** does.

CI fuel system: the **fuel injection pump** builds very high pressure (100–2000 bar) and the **injector (nozzle)** sprays very fine diesel droplets into the cylinder. In modern **common-rail (CRDI)** systems a shared rail is always kept at high pressure. **Ignition system (SI):** battery system (cars) or **magneto** (motorcycles, no battery); the ignition coil steps up 12 V to 20,000–30,000 V so that a spark jumps across the spark plug gap.

Cooling: combustion temperature exceeds ≈ 2000°C; if the cylinder is not cooled the lubricating oil will burn and the piston will seize. **Air cooling** — fins on the cylinder (motorcycles); **water cooling** — water jacket around the cylinder, pump, **radiator** and **thermostat** (stops water going to the radiator in a cold engine so it warms up quickly; opens at about 80°C).

Lubrication: reduces friction and wear, cools, seals and cleans. Systems: **mist (petroil)** — oil mixed with petrol in two-strokes; **splash** — a scoop on the connecting rod splashes oil (small engines); **pressure (forced)** — a pump sends oil to every bearing (cars/trucks); **dry sump** — racing/aircraft engines. **Viscosity index (VI)** — how little the oil's viscosity changes with temperature; higher VI is better.

1. Key Points

- **Indicated power (IP):** power developed by gases inside the cylinder; IP = p_mLAnk/60, where n = N/2 (4-stroke) or N (2-stroke), k = number of cylinders. p_m is obtained from the indicator diagram.
- **Brake power (BP):** useful power available at the crankshaft; BP = 2πNT/60. It is measured by a dynamometer (Prony brake, rope brake, hydraulic, electrical).
- **Friction power (FP):** FP = IP − BP; lost in piston-ring friction, bearings, driving accessories and pumping losses.
- **Mechanical efficiency:** η_m = BP/IP (usually 80–90%).
- **Thermal efficiency:** brake thermal η_{bth} = BP/(m_f × CV); indicated thermal η_{ith} = IP/(m_f × CV); η_{bth} = η_m × η_{ith}.
- **Volumetric efficiency:** η_v = actual volume of air drawn (at ambient conditions) ÷ swept volume; 70–90% for naturally aspirated engines; above 100% with supercharging.
- **Specific fuel consumption (SFC):** bsfc = m_f/BP (kg/kWh); the lower, the more efficient. η_{bth} = 3600/(bsfc × CV).
- **Methods to find friction power:** Willan's line (diesel), Morse test (multi-cylinder SI), motoring test, and IP − BP from indicator diagram.
- **Morse test:** cutting off one cylinder at a time, the drop in BP gives that cylinder's IP; IP_n = BP − BP_n.
- **Heat balance sheet:** total fuel heat = BP + heat to cooling water + heat in exhaust + unaccounted losses (roughly 30 : 30 : 30 : 10).

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Indicated power	**$IP = p_m L A n k / 60$**
Brake power	**$BP = 2\pi NT / 60$**
Friction power	**$FP = IP - BP$**
Mechanical efficiency	**$\eta_m = BP/IP$**
Brake thermal efficiency	**$\eta_{bth} = BP / (m_f \times CV)$**
Indicated thermal efficiency	**$\eta_{ith} = IP / (m_f \times CV)$**
Relation	**$\eta_{bth} = \eta_m \times \eta_{ith}$**
Volumetric efficiency	**$\eta_v = V_{actual} / V_s$**
bsfc	**m_f / BP (kg/kWh)**
Rope brake	**$BP = (W - S) \pi(D + d) N / 60$**
Morse test	**$IP_n = BP - BP_n$**

3. Detailed Explanation

- **Indicator diagram:** the actual p–V diagram obtained from an engine indicator; area ÷ length × spring constant = mean effective pressure.
- **Willan's line:** plotting fuel consumption against BP at constant speed and extrapolating to zero fuel gives FP on the negative BP axis; suitable only for CI engines.
- **Supercharging and turbocharging:** supplying air above atmospheric pressure to burn more fuel; power increases; a turbocharger is driven by exhaust gases.
- **Performance with load:** mechanical efficiency rises with load (FP nearly constant), while bsfc first falls and then rises near full load.
- **Power ratings:** 1 kW = 1.341 HP; 1 metric HP = 735.5 W.

4. Comparison Table — Indicated Power vs Brake Power

Basis	Indicated power (IP)	Brake power (BP)
Where developed	Inside cylinder	At crankshaft
Measured by	Indicator diagram	Dynamometer
Formula	p_mLAnk/60	2πNT/60
Comparison	Higher	Lower (IP − FP)

5. Solved Numerical Problems

Example 1. A four-stroke single-cylinder engine: p_m = 6 bar, bore 100 mm, stroke 120 mm, N = 1500 rpm. Find IP.

Solution:

1. $A = \pi/4 \times 0.1^2 = 7.854 \times 10^{-3}$ m²; n = 1500/2 = 750
2. $IP = 6 \times 10^5 \times 0.12 \times 7.854 \times 10^{-3} \times 750/60$
3. IP = 7069 W = **7.07 kW**

Example 2. An engine gives 50 N·m at 2000 rpm and IP = 13 kW. Find BP, FP and mechanical efficiency.

Solution:

1. BP = $2\pi \times 2000 \times 50/60$ = 10,472 W = **10.47 kW**
2. FP = 13 − 10.47 = **2.53 kW**
3. η_m = 10.47/13 = **80.5%**

Example 3. An engine uses 3 kg/h of fuel (CV = 42,000 kJ/kg) at 10 kW BP. Find bsfc and brake thermal efficiency.

Solution:

1. bsfc = 3/10 = **0.3 kg/kWh**
2. η_{bth} = 10/(3/3600 × 42000) = 10/35 = **28.6%**

Example 4. A four-cylinder engine has BP = 30 kW; with each cylinder cut out, BP = 21, 21.5, 20.8, 21.2 kW. Find total IP and mechanical efficiency.

Solution:

1. IP = (30 − 21) + (30 − 21.5) + (30 − 20.8) + (30 − 21.2) = 9 + 8.5 + 9.2 + 8.8
2. IP = **35.5 kW**
3. η_m = 30/35.5 = **84.5%**

Example 5. Swept volume 1 litre; actual air drawn 0.85 litre (ambient). Find volumetric efficiency.

Solution:

1. η_v = 0.85/1 = **85%**

6. Frequently Asked Facts

- η_m = BP/IP; FP = IP − BP.
- Morse test – multi-cylinder SI engines; Willan's line – CI engines.
- Lower bsfc = higher efficiency; η_{bth} = 3600/(bsfc × CV).
- In the IP formula for four-stroke, n = N/2.

7. Note (Confusing Facts)

- IP is always more than BP; the difference is friction power.

- Volumetric efficiency can exceed 100% with supercharging.
- n = N for two-stroke and N/2 for four-stroke — a common mistake.

Remember: One line: $\eta_{bth} = \eta_m \times \eta_{ith}$. Assertion–Reason: (A) Mechanical efficiency rises with load. (R) Friction power stays nearly constant while BP increases. — Both true; R explains A.

30 Most Important One-Liner Facts — Performance of IC Engines

1. Power developed inside cylinder?

— Indicated power (IP).

2. Power at crankshaft?

— Brake power (BP).

3. Friction power?

— IP − BP.

4. Mechanical efficiency?

— BP/IP.

5. IP formula?

— p_mLAnk/60.

6. n for four-stroke?

— N/2.

7. n for two-stroke?

— N.

8. BP formula?

— 2πNT/60.

9. Instrument for BP?

— Dynamometer.

10. To find IP?

— Indicator diagram.

11. Brake thermal efficiency?

— BP/(m_f × CV).

12. Relation of η_{bth}, η_m, η_{ith}?

— $\eta_{bth} = \eta_m \times \eta_{ith}$.

13. Volumetric efficiency?

— Actual volume ÷ swept volume.

14. Usual volumetric efficiency?

— 70–90%.

15. When η_v > 100%?

— Supercharging.

16. Unit of bsfc?

— kg/kWh.

17. bsfc vs efficiency?

— Inverse.

18. FP method for multi-cylinder SI?

— Morse test.

19. FP method for CI engine?

— Willan's line.

20. IP of one cylinder in Morse test?

— BP − BP_n.

21. BP share in heat balance?

— ≈ 30%.

22. Usual mechanical efficiency?

— 80–90%.

23. Exhaust-driven supercharger?

— Turbocharger.

24. η_m with load?

— Increases.

25. 1 kW = ? HP

— 1.341.

26. Rope brake BP?

— (W − S)π(D + d)N/60.

27. η_{bth} from bsfc?

— 3600/(bsfc × CV).

28. Types of dynamometer?

— Prony, rope, hydraulic, electrical.

29. Components of FP?

— Rubbing, pumping, accessories.

30. Cooling water share?

— ≈ 30%.

Practice MCQs — Performance of IC Engines

Q1. Mechanical efficiency is:

(a) IP/BP

(b) BP/IP

(c) FP/IP

(d) BP/FP

✔ Answer: (b) BP/IP

Explanation: η_m = BP/IP.

Q2. Friction power =

(a) IP + BP

(b) IP − BP

(c) BP − IP

(d) IP × BP

✔ Answer: (b) IP − BP

Explanation: FP = IP − BP.

Q3. In the IP formula for a four-stroke engine, n =

(a) N

(b) N/2

(c) 2N

(d) N/4

✔ Answer: (b) N/2

Explanation: One power stroke per 2 revolutions.

Q4. Test to find FP of a multi-cylinder petrol engine:

(a) Willan's line

(b) Morse test

(c) Retardation test

(d) Heat balance

✔ Answer: (b) Morse test

Explanation: Cutting each cylinder.

Q5. Willan's line method is suitable for:

(a) SI engines

(b) CI engines

(c) Gas turbines

(d) Steam turbines

✔ Answer: (b) CI engines

Explanation: Fuel vs BP is linear (CI).

Q6. Brake thermal efficiency =

(a) IP/(m_f CV)

(b) BP/(m_f CV)

(c) BP/IP

(d) m_f CV/BP

✔ Answer: (b) BP/(m_f CV)

Explanation: Useful power ÷ fuel energy.

Q7. IP = 20 kW, BP = 16 kW — mechanical efficiency:

(a) 75%
(b) 80%
(c) 85%
(d) 125%

✔ **Answer: (b) 80%**

Explanation: 16/20.

Q8. Volumetric efficiency can exceed 100% with:

(a) Supercharging
(b) Throttling
(c) High altitude
(d) Hot intake

✔ **Answer: (a) Supercharging**

Explanation: Pressurised air.

Q9. Unit of bsfc:

(a) kW/kg
(b) kg/kWh
(c) kJ/kg
(d) kg/s

✔ **Answer: (b) kg/kWh**

Explanation: Mass per energy.

Q10. BP is measured by a:

(a) Indicator
(b) Dynamometer
(c) Manometer
(d) Calorimeter

✔ **Answer: (b) Dynamometer**

Explanation: Torque measurement.

Q11. $\eta_m = 0.8$, $\eta_{ith} = 0.4$ — η_{bth}:

(a) 0.5
(b) 0.32
(c) 1.2
(d) 0.2

✔ **Answer: (b) 0.32**

Explanation: 0.8 × 0.4.

Q12. Mechanical efficiency as load increases:

(a) Decreases
(b) Increases
(c) Constant
(d) Zero

✔ **Answer: (b) Increases**

Explanation: FP nearly constant.

Q13. Share of BP in heat balance is about:

(a) 10%
(b) 30%
(c) 60%
(d) 90%

✔ **Answer: (b) 30%**

Explanation: ≈ 30%.

Q14. Device driven by exhaust gases:

(a) Supercharger
(b) Turbocharger
(c) Carburettor
(d) Governor

✔ **Answer: (b) Turbocharger**

Explanation: Turbocharger.

Q15. bsfc = 0.25 kg/kWh, CV = 40,000 kJ/kg — η_{bth}:

(a) 25%
(b) 36%
(c) 40%
(d) 45%

✔ **Answer: (b) 36%**

Explanation: 3600/(0.25 × 40000) = 0.36.

SSC JE / UPSSSC JE Previous Year Questions — Performance of IC Engines

Q1. Morse test is used to find:

SSC JE (Previous Year)

(a) BP
(b) IP of a multi-cylinder engine
(c) Fuel consumption
(d) Air consumption

✔ **Answer: (b) IP of a multi-cylinder engine**

Explanation: IP of each cylinder.

Q2. Friction power equals:

SSC JE (Previous Year)

(a) IP + BP
(b) IP − BP
(c) BP − IP
(d) IP/BP

✔ **Answer: (b) IP − BP**

Explanation: FP = IP − BP.

Q3. Formula of mechanical efficiency:

SSC JE (Previous Year)

(a) BP/IP
(b) IP/BP
(c) FP/BP
(d) BP/FP

✔ **Answer: (a) BP/IP**

Explanation: η_m = BP/IP.

Q4. A dynamometer measures:

SSC JE (Previous Year)

(a) Indicated power
(b) Brake power
(c) Friction power
(d) Fuel power

✔ **Answer: (b) Brake power**

Explanation: BP from torque.

Q5. The indicator diagram gives:

SSC JE (Previous Year)

(a) Brake power

(b) Mean effective pressure

(c) Friction

(d) Fuel consumption

✔ **Answer: (b) Mean effective pressure**

Explanation: p_m from area.

UPSSSC ALERT